

Skills Summary

Place-Value Synthesis Review

Use this reference while completing your worksheet or when studying.

Skill 1 — Expanded Form

Rule: expanded form writes a number as a sum of place values, largest place first. Going back to standard form, put a **zero** in any place with no addend.

Example: Write $90,000 + 1,000 + 50 + 3$ in standard form.

Step 1. Each addend fills one place; the hundreds place has no addend, so it gets a zero.

Step 2. 9 ten-thousands, 1 thousand, 0 hundreds, 5 tens, 3 ones: 91,053 $\Rightarrow$ **91,053**

Try it: Write $20,000 + 700 + 6$ in standard form.

check: 20,706

Skill 2 — Comparing Numbers

Rule: more digits wins; with the same number of digits, compare place by place from the left and let the first difference decide. Trailing digits never outrank a higher place.

Example: Compare 54,631 and 54,613.

Step 1. Same digit count; scan from the left: ten-thousands, thousands, and hundreds all tie.

Step 2. Tens differ: $3 > 1$, and everything after the tens no longer matters.

$\Rightarrow$ **54,631 > 54,613**

Try it: Compare 7,208 and 7,280 with $<$, $>$, or $=$.

check: 7,280 > 7,208

Skill 3 — Rounding Decisions

Rule: underline the digit in the named place; the digit one place right decides (0–4 down, 5–9 up); everything right of the named place becomes zero. A round-up of a 9 carries left.

Example: Round 6,285 to the nearest hundred.

Step 1. Underline the hundreds digit 2; the decider is the tens digit, 8.

Step 2. $8 \geq 5$, so 2 hundreds become 3 hundreds: $6,285 \rightarrow 6,300 \Rightarrow$ **6,300**

Try it: Round 6,285 to the nearest *thousand*.

check: 6,000

(!) Watch out: in a mixed review, half the battle is picking the tool. “Standard form” means build the number; a symbol blank means compare place-by-place; “nearest . . .” means round. Name the tool before you compute.